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Multiscale, multiwavelength extraction of sources and filaments using separation of the structural components: *getsf*

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ABSTRACT

High-quality astronomical images delivered by modern ground-based and space observatories demand adequate, reliable software for their analysis and accurate extraction of sources, filaments, and other structures, containing massive amounts of detailed information about the complex physical processes in space. The multiwavelength observations with highly variable angular resolutions across wavebands require extraction tools that preserve and use the invaluable high-resolution information. Complex fluctuating backgrounds and filamentary structures appear differently on various scales, calling for multiscale approaches for complete and reliable extraction of sources and filaments. The availability of many extraction tools with varying qualities highlights the need to use standard model benchmarks for choosing the most reliable and accurate method for astrophysical research. This paper presents *getsf*, a new method for extracting sources and filaments in astronomical images using separation of their structural components, designed to handle multiwavelength sets of images and very complex filamentary backgrounds. The method spatially decomposes the original images and separates the structural components of sources and filaments from each other and from their backgrounds, flattening their resulting images. It spatially decomposes the flattened components, combines them over wavelengths, detects the positions of sources and skeletons of filaments, and measures the detected sources and filaments, creating the output catalogs and images. The fully automated method has a single user-defined parameter (per image), the maximum size of the structures of interest to be extracted, that must be specified by users. This paper presents a realistic multiwavelength set of simulated benchmark images that can serve as the standard benchmark problem to evaluate qualities of source- and filament-extraction methods. This paper describes *hires*, an improved algorithm for the derivation of high-resolution surface densities from multiwavelength far-infrared *Herschel* images. The algorithm allows creating the surface densities with angular resolutions that reach 5.6″ when the 70 µm image is used. If the shortest-wavelength image is too noisy or cannot be used for other reasons, slightly lower resolutions of 6.8−11.3″ are available from the 100 or 160 µm images. These high resolutions are useful for detailed studies of the structural diversity in molecular clouds. The codes *getsf* and *hires* are illustrated by their applications to a variety of images obtained with ground-based and space telescopes from the X-ray domain to the millimeter wavelengths.

Key words. stars: formation – infrared: ISM – submillimeter: ISM – methods: data analysis – techniques: image processing – techniques: photometric

1. Introduction

Multiwavelength far-infrared and submillimeter dust continuum observations with the large space telescopes *Spitzer*, *Herschel*, and *Planck* in the past decades greatly increased the amount and improved the quality of the available data in various areas of astrophysical research. Observed images with diffraction-limited angular resolutions and high sensitivity reveal an impressive diversity of the enormously complex structures, covering orders of magnitude in intensities and spatial scales. The images feature foremost the bright fluctuating backgrounds, omnipresent filaments, and huge numbers of sources of different physical nature, all blended with each other, whose appearance and resolution are often markedly different at short and long wavelengths. The massive amount of information that is coded in the fine structure of the observed images must contain clues to the complex physical processes taking place in space, but these clues are extremely difficult to decipher. It is quite clear that the era of these high-quality data from space telescopes and large ground-based interferometers, such as the Atacama Large Millimeter/submillimeter Array (ALMA), requires more sophisticated tools for their accurate analysis and correct interpretation than those developed for the lower-quality images of the past. Adequate extraction methods must be explicitly designed for the multiwavelength imaging observations with highly dissimilar angular resolutions across wavebands. They must also be able to handle the bright filamentary backgrounds that vary on all spatial scales, whose fluctuation levels differ by several orders of magnitude across the observed images.

The source- and filament-extraction methods are growing in numbers. In the area of star formation, a new method was published every year or two within the seven-year period after the launch of *Herschel*. Rosolowsky et al. (2008) devised *dendrograms* to describe the hierarchical structure of clumps observed in the data cubes from molecular line observations. The method carries out topological analysis of image structures by isophotal contours at varying intensity levels and represents them graphically as a tree. Molinari et al. (2011) created *cutex* to extract sources in star-forming regions observed with *Herschel*. The method analyzes multidirectional second derivatives of the observed image to detect sources, and it measures them by fitting elliptical Gaussians on a planar background to their peaks. Men'shchikov et al. (2012) developed *getsources*, the multiwavelength source extraction method for the *Herschel* observations of star-forming regions. The method spatially decomposes images, combines them into wavelength-independent detection images, subtracts the backgrounds of detected sources, and measures the sources, deblending them when they overlap. Kirk et al. (2013) presented *csar* for the *Herschel* images. The method analyzes areas of connected pixels that are bound by closed isophotal contours, descending to a predefined background level and partitioning peaks at their lowest isolated contours into sources. Berry (2015) created *fellwalker* to identify clumps in submillimeter data cubes. The method finds image peaks by tracing the line of the steepest ascent and identifies sources as the hill with the highest value found for all pixels in its neighborhood. Sousbie (2011) produced *disperse* to identify structures in the large-scale distribution of matter in the Universe. The method applies the computational topology to trace filaments and other structures. Men'shchikov (2013) developed *getfilaments* to improve the source extraction with *getsources* on the filamentary backgrounds observed with *Herschel*. The method separates filaments from sources in spatially decomposed images and subtracts them from the detection images, thereby reducing the rate of spurious sources. Schisano et al. (2014) devised a Hessian matrix-based approach to extract filaments in *Herschel* observations of the Galactic plane. The method analyzes multidimensional second derivatives to identify filaments and determine their properties. Clark et al. (2014) presented *rht* to characterize fibers in the interstellar H I medium. The method has been applied to various observations of diffuse H I, revealing alignment of the fibers along magnetic fields. Koch & Rosolowsky (2015) published *filfinder* to identify filaments in the *Herschel* images of star-forming regions. the method applies a mathematical morphology approach to isolating filaments in observed images. Juvela (2016) presented *tm* to trace filaments in observed images. The method matches a predefined template (stencil) of an elongated structure at each pixel of an image by shifting and rotating the template and analyzing the parameters of the matches.

These extraction methods all have several important issues. Sources and filaments are handled completely independently by these methods, although numerous *Herschel* observations have demonstrated that there is a tight physical relation between them. Most sources are found in filamentary structures, and the corresponding starless, prestellar, and protostellar cores are thought to form inside the structures that are created by dynamical processes, magnetic fields, and gravity within a molecular cloud. All major structural components of the observed images, that is, the background cloud, filamentary structures, and sources, are heavily blended with each other; curved filaments are even blended with themselves. The degree of their blending increases at longer wavelengths with lower angular resolutions, which increases the inaccuracies in their detections, measurements, and interpretations.

Most of the extraction methods focus on detecting structures, whereas the most important and difficult problem is measuring them accurately. Numerous algorithms only partition the image between sources and do not allow them to overlap, although deblending of the mixed emission of the structural components is an indispensable property of an accurate extraction method. For best detection and measurement results, source-extraction methods must be able to separate underlying filamentary structures and filament-extraction methods must be able to separate sources. The existing source- and filament-extraction methods use completely different approaches, and the quality of their results is expected to be very dissimilar.

It seems unlikely that methods that are based on very different approaches would give consistent results in terms of detection completeness, number of false-positive detections, and measurement accuracy. In practice, various methods do perform very differently, as can be shown quantitatively on simulated benchmarks for which the properties of all components are known. This highlights the need of systematic comparisons of different methods in order to understand their qualities, inaccuracies, and biases. The danger is real that numerous uncalibrated methods are applied for the same type of star formation studies, which would give inconsistent, contradictory results and incorrect conclusions. This would create serious, lasting problems for the science.

Source- and filament-extraction methods are the critically important tools that must be calibrated and validated using a standard set of benchmark images with fully known properties of all components before they are applied in astrophysical contexts. It would be desirable to use the same extraction tool to exclude any biases or dissimilarities that are caused by different methods. If a new extraction method is to be used, it must be tested on standard benchmarks to ensure that its detection and measurement qualities are consistent or better. This approach is usually practiced within research consortia, but this does not solve the global problem that the results obtained from the same data by different consortia or research groups using different tools may still be affected by the uncalibrated (or suboptimal) tools that were used.

This paper presents *getsf*, a new multiwavelength method for extracting sources and filaments. It also describes a realistic simulated benchmark, resembling the *Herschel* images of star-forming regions, which is used below to illustrate the method and in a separate paper (Men'shchikov 2021) to quantitatively evaluate its performance. The multiwavelength benchmark simulates the images of a dense cloud with strong nonuniform fluctuations, a wide dense filament with a power-law intensity profile, and hundreds of radiative transfer models of starless and protostellar cores with wide ranges of sizes, masses, and profiles. The simulated benchmark with fully known parameters allows quantitative analyses of extraction results and conclusive comparisons of different methods by evaluating their extraction completeness, reliability, and goodness, along with the detection and measurement accuracies. The multiwavelength images can serve as the standard benchmark problem for other source- and filament-extraction methods, allowing researchers to perform their own tests and choose the most reliable and accurate extraction method for their studies. Instead of publishing benchmarking results for some of the existing methods, it seems a better idea to provide researchers with the benchmark[1] and a quality evaluation system (Men'shchikov 2021) to enable comparisons of the methods of their choice. In practice, this approach of having own experience is much more convincing and it allows a consistent evaluation of newly developed methods.

The new source and filament extraction method *getsf* represents a major improvement over the previous algorithms *getsources*, *getfilaments*, and *getimages* (Men'shchikov et al. 2012; Men'shchikov 2013, 2017, hereafter referred to as Papers I, II, and III); throughout this paper, the three predecessors are collectively referred to as *getold*. The new method (Fig. 1) consistently handles two types of structures, sources and filaments, that are important for studies of star formation, separating the structural components from each other and from their backgrounds. All major processing steps of *getsf* employ spatial decomposition of images into a number of finely spaced

[1] `http://irfu.cea.fr/Pisp/alexander.menshchikov/#bench`

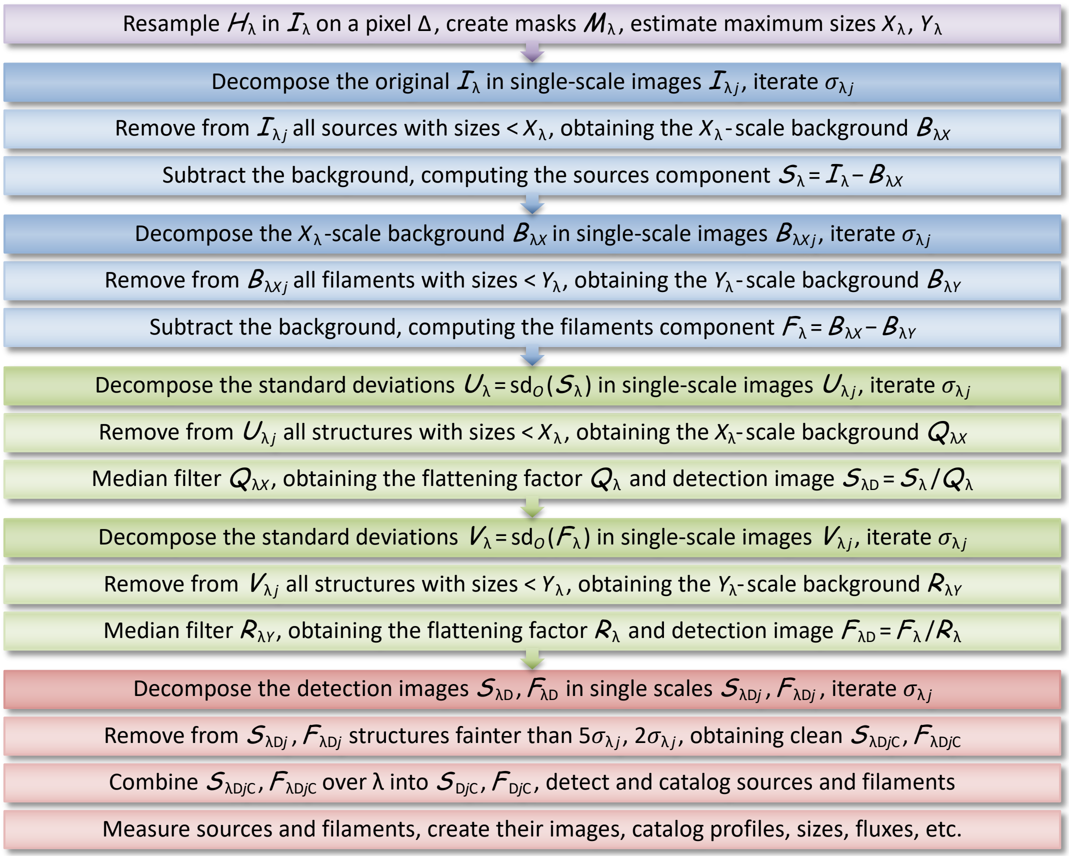

Fig. 1. Flowchart of the image processing steps in *getsf*. The colored blocks represent preparation (purple), background subtraction (blue), image flattening (green), and extraction of sources and filaments (red).

single-scale images to better isolate the contributions of structures with various widths. The method produces accurately flattened detection images with uniform levels of the residual background and noise fluctuations. To detect sources and filaments, *getsf* combines independent information contained in the multiwaveband single-scale images of the structural components, preserving the higher angular resolutions. Then *getsf* measures and catalogs the detected sources and filaments in their background-subtracted images. The fully automated method needs only one user-defined parameter, the maximum size of the structures of interest to extract, constrained by users from the input images on the basis of their research interests.

This work follows Papers I–III in advocating a clear distinction between the words *source* and *object*, unlike many publications in which it is implicitly assumed that the two are completely equivalent. "Source" is used in the context of the source extractions and statistical analysis of their results, and "object" is only used in the context of the physical interpretation of the extracted sources. In this paper, the sources are defined as the emission peaks (mostly unresolved) that are significantly stronger than the local surrounding fluctuations, indicating the presence of the physical objects in space that produced the observed emission. The implicit assumption that an unresolved far-infrared source on a complex fluctuating background contains emission of just one single object is invalid in general. Too often, an emission peak is actually a blend of many components, produced by different physical entities. This is illustrated by the recent images of the massive star-forming cloud W43-MM1 (Motte et al. 2018), obtained with the ALMA interferometer. This object appears as a single source in the *Herschel* images, even with the 5.6 and 11.3″ resolutions at 70 and 160 μm. However, the ALMA image (Sect. 4.8) displays a rich cluster of much smaller sources that are unresolved or just slightly resolved even at the 0.44″ resolution.

Section 2 describes the new multiwavelength benchmark for source- and filament-extraction methods, resembling the *Herschel* observations of star-forming regions. Section 3 presents *getsf*, the new source- and filament-extraction method, employing separation of the structural components. Section 4 illustrates the performance of *getsf* on a large variety of images that were obtained with different telescopes in a wide spectral range, from X-rays to millimeter wavelengths. Section 5 describes all strengths and limitations of *getsf*. Section 6 presents a summary of this work. Appendix A discusses inaccuracies of the surface densities and temperatures, derived by spectral fitting of the images. Appendix B describes the single-scale spatial decomposition that is used by *getsf* in its processing steps. Appendix C gives details on the software.

In this paper, images are represented by capital calligraphic characters (e.g., $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$) and software names and numerical methods are typeset slanted (e.g., *getsf*) to distinguish them from other emphasized words. The curly brackets {} are used to collectively refer to either of the characters, separated by vertical lines. For example, $\{a|b\}$ refers to a or b and $\{A|B\}_{\{a|b\}c}$ expands to $A_{\{a|b\}c}$ or $B_{\{a|b\}c}$, as well as to A_{ac}, A_{bc}, B_{ac}, or B_{bc}.

2. Benchmark for source and filament extractions

Realistic multiwavelength, multicomponent images of a simulated star-forming region were computed to present *getsf* in this paper and to compare its performance with the previous

Fig. 2. Background surface densities ($\mathcal{D}_\mathrm{B}, \mathcal{D}_\mathrm{C}$) and average line-of-sight dust temperatures ($\mathcal{T}_\mathrm{C}$) used to compute the simulated *Herschel* images C_λ of the filamentary cloud from Eq. (4). Square-root color mapping.

benchmark that was used in Papers I and III. The benchmark images were created for all *Herschel* wavebands (at λ of 70, 100, 160, 250, 350, and 500 µm). They consist of independent structural components: a background cloud $\mathcal{B}_\lambda$, a long filament $\mathcal{F}_\lambda$, round sources $\mathcal{S}_\lambda$, and small-scale instrumental noise $\mathcal{N}_\lambda$:

$$\mathcal{H}_\lambda = \mathcal{B}_\lambda + \mathcal{F}_\lambda + \mathcal{S}_\lambda + \mathcal{N}_\lambda = \mathcal{C}_\lambda + \mathcal{S}_\lambda + \mathcal{N}_\lambda, \tag{1}$$

where $\mathcal{C}_\lambda = \mathcal{B}_\lambda + \mathcal{F}_\lambda$ is the emission intensity of the filamentary background. All simulated images were computed on a 2″ pixel grid with 2690 × 2690 pixels, covering 1.5° × 1.5° or 3.7 pc at a distance D = 140 pc of the nearest star-forming regions (e.g., those in Taurus or Ophiuchus).

2.1. Simulated filamentary background

An image of the background surface density was computed from a purely synthetic scale-free background $\mathcal{D}_\mathrm{A}$ (cf. Paper I), with $N_{\mathrm{H}_2} \sim 2.7 \times 10^{20}$ to $5 \times 10^{22}\,\mathrm{cm}^{-2}$ that had uniform fluctuations across the entire image. To simulate complex astrophysical backgrounds with strongly nonuniform fluctuations (e.g., Könyves et al. 2015), $\mathcal{D}_\mathrm{A}$ was multiplied by a circular shape $\mathcal{P}$ with a radial profile defined by Eq. (2) below (with $\Theta = 1500''$ and $\zeta = 2$), normalized to unity and centered on the image; finally, a constant value of $1.5 \times 10^{21}\,\mathrm{cm}^{-2}$ was added to increase the minimum value. The surface densities of the resulting background cloud image $\mathcal{D}_\mathrm{B}$ (Fig. 2) are 1.5×10^{21} to $4.8 \times 10^{22}\,\mathrm{cm}^{-2}$ and the fluctuations differ by approximately two orders of magnitude. The total mass of the cloud is $M_\mathrm{B} = 1.78 \times 10^3\,M_\odot$.

To simulate filamentary backgrounds, a long spiral filament was added to the background cloud $\mathcal{D}_\mathrm{B}$. The spiral shape was chosen so that the filament occupied various areas of the cloud with very different surface densities and to cause the filament to be blended (with itself) to some extent. The spiral filament image $\mathcal{D}_\mathrm{F}$ has a crest value of $N_0 = 10^{23}\,\mathrm{cm}^{-2}$, a full width at half-maximum (FWHM) $W = 0.1$ pc, and a radial profile similar to those observed with *Herschel* in star-forming regions (e.g., Arzoumanian et al. 2011, 2019),

$$N_{\mathrm{H}_2}(\theta) = N_0 \left(1 + (2^{1/\zeta} - 1)\,(\theta/\Theta)^2\right)^{-\zeta}, \tag{2}$$

where θ is the angular distance, Θ is the structure half-width at half-maximum, and ζ is a power-law exponent. With $\Theta = 75''$ (or 0.05 pc at D = 140 pc) and $\zeta = 1.5$, this Moffat (Plummer) function approximates a Gaussian of 0.1 pc (FWHM) in its core and it transforms into a power-law profile $N_{\mathrm{H}_2}(\theta) \propto \theta^{-3}$ for $\theta \gg \Theta$. The filament mass $M_\mathrm{F} = 3.04 \times 10^3\,M_\odot$ and length $L_\mathrm{F} = 10.5$ pc correspond to the linear density $\Lambda_\mathrm{F} = 290\,M_\odot\,\mathrm{pc}^{-1}$. The resulting surface densities $\mathcal{D}_\mathrm{C} = \mathcal{D}_\mathrm{B} + \mathcal{D}_\mathrm{F}$ of the filamentary cloud are in the range of 1.7×10^{21} to $1.4 \times 10^{23}\,\mathrm{cm}^{-2}$ (Fig. 2), and its total mass is $M_\mathrm{C} = 4.82 \times 10^3\,M_\odot$.

To approximate the nonuniform line-of-sight dust temperatures of the star-forming clouds observed with *Herschel* (e.g., Men'shchikov et al. 2010; Arzoumanian et al. 2019), an image of average line-of-sight temperatures was improvised as

$$\mathcal{T}_\mathrm{C} = 200\left(10^{-20}\mathcal{D}_\mathrm{C} + 20\right)^{-1} + 15\,\mathrm{K}. \tag{3}$$

The pixel values of the resulting temperature image $\mathcal{T}_\mathrm{C}$ range between 15 K in the innermost areas of the filamentary cloud and 20 K in its outermost parts (Fig. 2). The temperatures from Eq. (3) were used to simulate the cloud images $\mathcal{C}_\lambda$ in all *Herschel* wavebands, assuming optically thin dust emission:

$$\mathcal{C}_\nu = B_\nu(\mathcal{T}_\mathrm{C})\,\mathcal{D}_\mathrm{C}\,\kappa_\nu \eta \mu m_\mathrm{H}, \tag{4}$$

where B_ν is the blackbody intensity, κ_ν is the dust opacity, η = 0.01 is the dust-to-gas mass ratio, μ = 2.8 is the mean molecular weight per H_2 molecule, and m_H is the hydrogen mass. The dust opacity was parameterized as a power law $\kappa_\nu = \kappa_0\,(\nu/\nu_0)^\beta$ with $\kappa_0 = 9.31\,\mathrm{cm}^2\,\mathrm{g}^{-1}$ (per gram of dust), $\lambda_0 = 300$ µm, and $\beta = 2$.

2.2. Simulated starless and protostellar cores

To populate the filamentary cloud with realistic sources, 156 radiative transfer models were computed by a numerical solution of the dust continuum radiative transfer problem in spherical geometry (using *modust*, Bouwman 2001). The models adopted tabulated absorption opacities κ_abs for dust grains with thin ice mantles (Ossenkopf & Henning 1994), corresponding to a density $n_\mathrm{H} = 10^6\,\mathrm{cm}^{-3}$ and coagulation time $t = 10^5$ yr. The opacity values at $\lambda > 160$ µm were replaced with a power law $\kappa_\lambda \propto \lambda^{-2}$, consistent with the parameterization used in Eq. (4).

The models of three populations of starless cores and one population of protostellar cores cover wide ranges of masses (from 0.05 to 2 $M_\odot$) and half-maximum sizes (from ~0.001 to

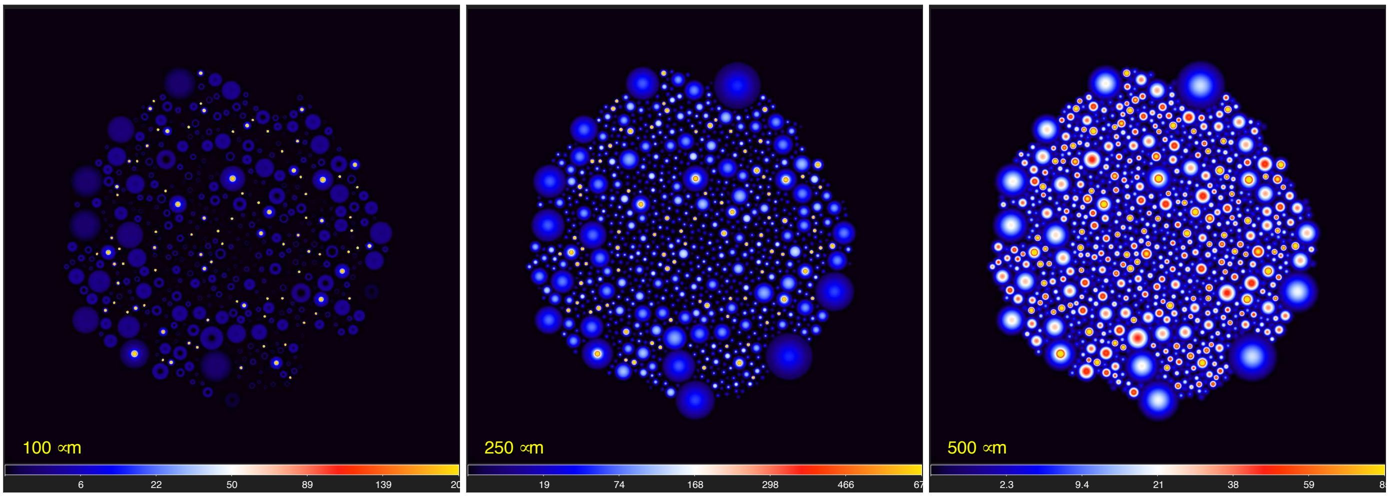

Fig. 3. Component of sources $\mathcal{S}_{\lambda}$ that is composed of the images of radiative transfer models of 828 starless and 91 protostellar cores and convolved to the *Herschel* resolutions O_{λ} (cf. Sect. 1), shown at three selected wavelengths. Only the bright unresolved emission peaks of the protostellar cores, clearly visible at 100 µm, appear in the 70 µm image (not shown). Square-root color mapping.

0.1 pc). Density profiles of the critical Bonnor–Ebert spheres were adopted for starless cores, whereas the protostellar cores have power-law densities $\rho(r) \propto r^{-2}$. Starless cores consist of low-, medium-, and high-density subpopulations, following the $M \propto R$ relation for the isothermal Bonnor–Ebert spheres (with $T_{\rm BE}$ = 7, 14, 28 K) in the area of the mass–radius diagram occupied by prestellar cores observed in the Ophiuchus and Orion star-forming regions (Motte et al. 1998, 2001).

Both types of cores were embedded in background spherical clouds with a uniform surface density of 3×10^{21} cm^{-2} and outer radius of 1.4×10^{5} AU (1000″ or 0.68 pc). In an isotropic interstellar radiation field (Black 1994) with the strength parameter $G_0 = 10$ (e.g., Parravano et al. 2003), the embedding clouds acquired temperatures of $T \approx 22$ K at their edges, consistent with the highest values of $\mathcal{T}_{\rm C}$ from Eq. (3). The embedding clouds lowered $T(r)$ toward the interiors of both starless and protostellar cores. Accreting protostars in the centers of the protostellar cores, however, produced luminosity $L_{\rm A} \propto M$ and thus sharply peaked temperature distributions deeper in their central parts.

2.3. Complete simulated images

Individual surface density images of the models of 828 starless and 91 protostellar cores were distributed in the dense areas ($N_{\rm H_2} \geq 5 \times 10^{21}$ cm^{-2}) of the filamentary cloud $\mathcal{D}_{\rm C}$. They were added quasi-randomly, without overlapping, to the $\mathcal{D}_{\rm C}$ image at positions, where their peak surface density exceeded that of the cloud $N_{\rm H_2}$ value. An initial mass function (IMF)-like power-law mass function with a slope dN/dM of −1.7 was used to determine the numbers of models per mass bin $\delta \log_{10} M \approx 0.1$ in each of the four populations. This resulted in the surface densities $\mathcal{D}_{\rm S}$, the intensities $\mathcal{S}_{\lambda}$ of sources (Fig. 3), and in the complete simulated images $\mathcal{C}_{\lambda} + \mathcal{S}_{\lambda}$.

The final simulated *Herschel* images $\mathcal{H}_{\lambda}$ from Eq. (1) of the modeled star-forming region were obtained by adding different realizations of the random Gaussian noise $\mathcal{N}_{\lambda}$ at 70, 100, 160, 250, 350, and 500 µm and convolving the resulting images to the angular resolutions O_{λ} of 8.4, 9.4, 13.5, 18.2, 24.9, and 36.3″, respectively (Fig. 4). The resulting images $\mathcal{H}_{\lambda}$ have σ noise levels of 6, 6, 5.5, 2.5, 1.2, and 0.5 MJy sr^{-1}, resembling the actual noise measured in the *Herschel* images of the Rosette molecular complex (Motte et al. 2010).

3. Source- and filament-extraction method

The main processing steps of *getsf* are outlined in Fig. 1, where several major blocks of the algorithm are highlighted. The method may be summarized as follows: (1) preparation of a complete set of images for an extraction, (2) separation of the structural components of sources and filaments from their backgrounds, (3) flattening of the residual noise and background fluctuations in the images of sources and filaments, (4) combination of the flattened components of sources and filaments over selected wavebands, (5) detection of sources and filaments in the combined images of the components, and (6) measurements of the properties of the detected sources and filaments.

Like its predecessors, *getsf* has just a single, user-definable parameter: the maximum size (width) of the structures of interest to extract. Internal parameters of *getsf* have been carefully calibrated and verified in numerous tests using large numbers of diverse images (both simulated and real-life observed images) to ensure that *getsf* works in all cases. This approach rests on the conviction that high-quality extraction methods for scientific applications must not depend on the human factor. It is the responsibility of the creator of a numerical method to make it as general as possible and to minimize the number of free parameters as much as possible. An internal multidimensional parameter space of complex numerical tools must never be delegated to the end user to explore if the aim is to obtain consistent and reliable scientific results.

3.1. Preparation of images for extraction

The multiwavelength extraction methods must be able to use all available information contained in the observed images across various wavebands with different angular resolutions. It is usually beneficial to collect all available images for a specific region of the sky under study.

3.1.1. Original observed set of images

To prepare multiwavelength $\mathcal{H}_{\lambda}$ for processing with *getsf*, it is necessary to convert them into the images $\mathcal{I}_{\lambda}$, all on the same grid of pixels. To this end, *getsf* resamples all images (using *swarp*, Bertin et al. 2002) on a pixel size, chosen to be

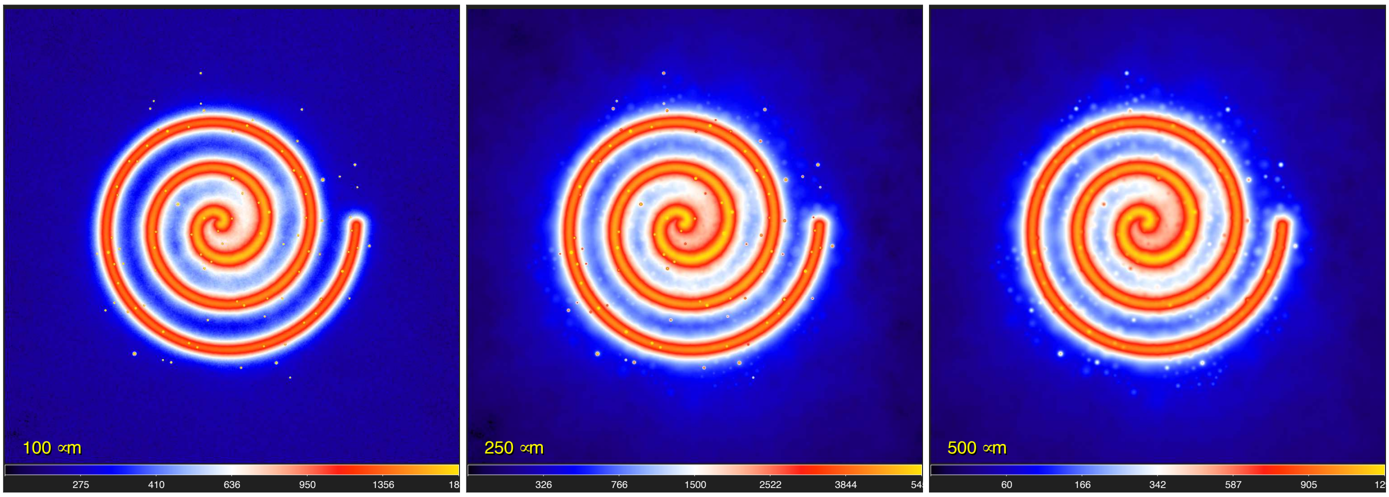

Fig. 4. Images $\mathcal{H}_\lambda$ of the simulated star-forming region, defined by Eq. (1)), shown at three selected wavelengths. The benchmark images are a superposition of four structural components: the background $\mathcal{B}_\lambda$, the filament $\mathcal{F}_\lambda$, the sources $\mathcal{S}_\lambda$, and the noise $\mathcal{N}_\lambda$. Two simpler variants of this benchmark are also available: without the filament and without the background. Square-root color mapping.

optimal for the highest-resolution images available. It is very important to carefully verify alignment of the resampled $\mathcal{I}_\lambda$ and correct it (if necessary) to ensure that all unresolved intensity peaks remain on the same pixel across all wavebands. To reveal possible misalignments, it is sufficient to open each pair of prepared images in *ds9* (Joye & Mandel 2003) and blink the two frames, going from the highest-resolution to the lowest-resolution images.

Most astronomical images have irregularly shaped coverage and limited usable areas that differ between wavebands. To include only the "good" parts of the $\mathcal{I}_\lambda$ coverage in the image processing, it is necessary to create masks $\mathcal{M}_\lambda$ (with pixel values 1 or 0). With these masks, *getsf* can process only the good areas of $\mathcal{I}_\lambda$ that have a mask value of 1. To facilitate the image preparation, *getsf* always creates default masks $\mathcal{M}_\lambda = 1$. However, for most real observations, the masks must be prepared very carefully and independently for each image. To manually create the masks, one can use *imagej* (Abràmoff et al. 2004) or *gimp*[2] that allows users to create a polygon over an image, convert the polygon into a mask, and save it in the FITS format.

3.1.2. Derived high-resolution images

The multiwavelength far-infrared *Herschel* images open the possibility of computing maps of surface density and dust temperature by fitting the spectral shapes Π_λ of the image pixels. The standard procedure assumes that (1) the original images represent optically thin thermal emission of dust grains with a power-law opacity $\kappa_\lambda \propto \lambda^{-\beta}$ and a constant β value, (2) the dust temperature is constant along the lines of sight passing through each pixel of the images, and (3) the lines of sight are not contaminated by unrelated radiation at either end, in front of the observed structures and behind them. Unfortunately, one or more of the assumptions are likely to be invalid, especially the stipulation of the opacity law and the constant line-of-sight temperatures (e.g., Men'shchikov 2016). The values of the derived surface densities and temperatures therefore must be considered as fairly unreliable and implying large error bars.

When we assume that the observations include the *Herschel* images, the spectral shapes Π_λ of each pixel can be fit at several wavelengths (160−500 µm) and resolutions (18.2–36.3″), which results in three sets of surface densities and dust temperatures. The highest-resolution derived images are the least reliable because they are obtained from fitting only two images (at 160 and 250 µm), whereas the lowest-resolution maps are the most accurate because they come from fitting four independent images (at 160, 250, 350, and 500 µm).

In an attempt to combine the higher accuracy of the lower-resolution images with the higher angular resolutions of the less accurate images, Palmeirim et al. (2013) published a simple algorithm that uses complementary spatial information contained in the observed images to create a surface density image with the resolution $O_{\rm P} = 18.2''$ of the 250 µm image. When this approach is extended to temperatures, the sharper images can be computed by adding the higher-resolution information to the low-resolution images as differential terms,

$$\{\mathcal{D}|\mathcal{T}\}_{\rm P} = \{\mathcal{D}|\mathcal{T}\}_4 + \delta\{\mathcal{D}|\mathcal{T}\}_3 + \delta\{\mathcal{D}|\mathcal{T}\}_2, \tag{5}$$

where the base surface density and temperature $\{\mathcal{D}|\mathcal{T}\}_4$ are derived by fitting the 160, 250, 350, and 500 µm images at the lowest resolution $O_{500} = 36.3''$. The additional terms, containing the higher-resolution contributions, are produced by unsharp masking,

$$\delta\{\mathcal{D}|\mathcal{T}\}_{\{2|3\}} = \{\mathcal{D}|\mathcal{T}\}_{\{2|3\}} - \mathcal{G}_{\{3|4\}} * \{\mathcal{D}|\mathcal{T}\}_{\{2|3\}}, \tag{6}$$

where $\{\mathcal{D}|\mathcal{T}\}_3$ are computed by fitting the three images at 160, 250, and 350 µm at the resolution $O_{350} = 24.9''$, and $\{\mathcal{D}|\mathcal{T}\}_2$ are obtained by fitting the two images at 160 and 250 µm at the resolution $O_{250} = 18.2''$; the Gaussian kernels $\mathcal{G}_{\{3|4\}}$ convolve the images to the next lower resolutions $O_{\{350|500\}}$.

The following generalization of the above algorithm allows deriving surface densities and temperatures with any (arbitrarily high) angular resolution existing among the observed $\mathcal{I}_\lambda$. The three independently derived maps of temperatures $\mathcal{T}_{\{2|3|4\}}$ with the resolutions of 18.2–36.3″ and six observed *Herschel* images with their native resolutions O_λ of 8.4–36.3″ define 18 surface densities,

$$\mathcal{D}_{O_\lambda\{2|3|4\}} = \frac{\mathcal{I}_\nu}{B_\nu(\mathcal{T}_{\{2|3|4\}})\,\kappa_\nu \eta \mu m_{\rm H}}, \tag{7}$$

with the assumptions and parameterizations of Eq. (4). It is required that the resolution of temperatures must not be higher

[2] http://www.gimp.org/

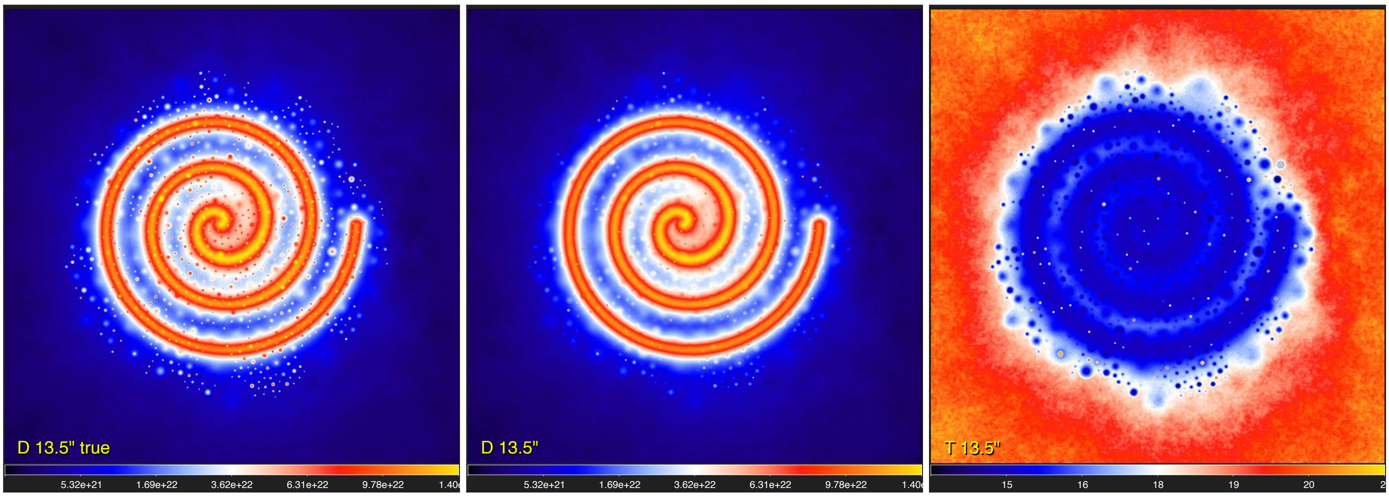

Fig. 5. Derived surface densities and temperatures (Sect. 3.1.2). The true model image $\mathcal{D}_{\rm C} + \mathcal{D}_{\rm S}$ and the *hires* surface density $\mathcal{D}_{13''}$ and temperature $\mathcal{T}_{13''}$ derived from Eq. (8) with $\lambda_{\rm H} = 160\,\mu$m ($O_{\rm H} = 13.5''$) are shown. Many of the sources, clearly visible in the true image (*left*), are not discernible in the derived surface density (*middle*) because of the inaccuracies in the temperatures from fitting spectral shapes Π_λ. Square-root color mapping, except the *right panel* with linear mapping.

than O_λ, which excludes $\mathcal{D}_{O_{350}2}$ and $\mathcal{D}_{O_{500}\{2|3\}}$ from the algorithm and provides 15 independently computed variants of the surface densities of the observed region, with different resolutions. The high-resolution surface density image is computed as

$$\mathcal{D}_{O_{\rm H}} = \mathcal{D}_{O_{500}} + \sum_{\lambda=\lambda_{\rm H}}^{500} \max\left(\delta\mathcal{D}_{O_\lambda 2}, \delta\mathcal{D}_{O_\lambda 3}, \delta\mathcal{D}_{O_\lambda 4}\right), \tag{8}$$

where $\lambda_{\rm H}$ denotes the wavelength of the image $\mathcal{I}_{\lambda_{\rm H}}$ with the desired angular resolution $O_{\rm H} \equiv O_{\lambda_{\rm H}}$ and the differential terms with higher-resolution information are obtained by the same unsharp masking,

$$\delta\mathcal{D}_{O_\lambda\{2|3|4\}} = \mathcal{D}_{O_\lambda\{2|3|4\}} - \mathcal{G}_{O_{\lambda+}} * \mathcal{D}_{O_\lambda\{2|3|4\}}, \tag{9}$$

where $\mathcal{G}_{O_{\lambda+}}$ is the Gaussian kernel (regarded as the delta function at $500\,\mu$m), convolving $\mathcal{D}_{O_\lambda\{2|3|4\}}$ to a lower resolution of the next longer wavelength. For images at $\lambda < 250\,\mu$m, only the positive values of $\delta\mathcal{D}_{O_\lambda\{2|3|4\}}$ are used in Eq. (8) to circumvent the problem of creating artificial depressions and negative pixels around strong peaks due to the resolution mismatch ($O_\lambda < O_{250}$) between $\mathcal{I}_\lambda$ and the lower-resolution $\mathcal{T}_{\{2|3|4\}}$ in Eq. (7).

The problem is caused by the sharp radial temperature gradients toward the unresolved centers of protostellar cores (cf. Men'shchikov 2016). They are smeared out by the low resolutions of 18.2–36.3″, hence the fitting of Π_λ leads to underestimated temperatures $\mathcal{T}_{\{2|3|4\}}$ and overestimated values (within an order of magnitude) of peak surface densities at higher resolutions $O_\lambda < O_{250}$. This means that unsharp masking of the overestimated peaks could create negative annuli in $\delta\mathcal{D}_{O_\lambda\{2|3|4\}}$ and negative pixels in $\mathcal{D}_{O_{\rm H}}$. Fortunately, the surface densities are quite accurate outside the unresolved peaks (Appendix A).

A slight modification of Eq. (8) allows deriving the high-resolution surface densities with an enhanced contrast of all unresolved or slightly resolved structures,

$$\mathcal{D}^{+}_{O_{\rm H}} = \mathcal{D}_{O_{\rm H}} + \sum_{\lambda=\lambda_{\rm H}}^{500} \sum_{n=2}^{4} \max\left(\delta\mathcal{D}_{O_\lambda n}, 0\right), \tag{10}$$

where the positive parts of the differential high-resolution terms from Eq. (9) are added to $\mathcal{D}_{O_{\rm H}}$. These high-contrast images may be useful for detection of unresolved structures because the latter are usually diluted by the observations with insufficient angular resolution; a higher contrast improves their visibility.

A high-resolution temperature $\mathcal{T}_{O_{\rm H}}$, consistent with the high-resolution surface density $\mathcal{D}_{O_{\rm H}}$, is computed by numerically inverting the Planck function,

$$\mathcal{T}_{O_{\rm H}} = B^{-1}_{\nu_{\rm H}}\left(\frac{\mathcal{I}_{\nu_{\rm H}}}{\mathcal{D}_{O_{\rm H}}\kappa_{\nu_{\rm H}}\eta\mu m_{\rm H}}\right), \tag{11}$$

with $\nu_{\rm H} = c\lambda^{-1}_{\rm H}$, where c is the speed of light. The high-resolution images $\{\mathcal{D}|\mathcal{T}\}_{13''}$ are shown in Fig. 5 along with the true simulated $\mathcal{D}_{\rm C} + \mathcal{D}_{\rm S}$ (Sect. 2.3). A comparison demonstrates that the pixel-fitting procedure reduces visibility of many unresolved or slightly resolved starless cores, which is the manifestation of the invalid assumption of the uniform line-of-sight temperatures. Starless (prestellar) cores have lower temperatures in their centers, and their smearing by an insufficient resolution leads to overestimated temperatures that suppress the surface density peaks (cf. Fig. A.1).

The new *hires* algorithm, outlined by Eqs. (7)–(11), brings the benefits of a resolution $O_{\rm H} \approx 8''$, twice better than $O_{\rm P}$ and four times better than O_{500}, if the image quality at the shortest wavelengths permits this. Moreover, the angular resolutions of the *Herschel* images at 70, 100, and $160\,\mu$m, obtained with a slow scanning speed of $20''\,{\rm s}^{-1}$, are even higher: 6, 7, and 11″, respectively. These observations, illustrated in Fig. 6, allow deriving the surface densities and temperatures with $O_{\rm H} \approx 6''$, a three times better resolution than when using Eq. (5). If the $70\,\mu$m image is too noisy or there is evidence of its strong contamination by emission unrelated to that of the adopted dust grains (e.g., polycyclic aromatic hydrocarbons or transiently heated very small dust grains), then the derived images may still have the 7–13″ resolution of the 100 or $160\,\mu$m wavebands. In addition to the high resolution, the images from Eq. (8) also have a better quality than those from Eq. (5) because they accumulate all available high-resolution information from the (up to 15) independently computed images $\mathcal{D}_{O_\lambda\{2|3|4\}}$ that use all three temperatures $\mathcal{T}_{\{2|3|4\}}$ with each original $\mathcal{I}_\lambda$.

The *hires* algorithm works with any number $2 \leq N \leq 6$ of *Herschel* wavebands. If the $160\,\mu$m image is unavailable or disabled, then the temperature $\mathcal{T}_2$ at the resolution O_{250} is